Science, Business and Innovation

Vrije Universiteit Amsterdam - Faculty of Science - Science, Business and Innovation - 2025-2026

The master program Science, Business and Innovation (SBI) aims to contribute to the persistent and growing need in academia, society and business for scientific and practical insights into valorization of scientific knowledge and findings. A combined understanding of the natural sciences, as well as of business and social sciences is supportive to knowledge and practice development in the field of valorization of (natural) scientific knowledge. Are you curious about how to integrate both the scientific and the business side of innovations? Do you want to play a crucial part in determining the economic success of innovations? Then the SBI Master at VU Amsterdam is the program for you.

The Master is a two years program and is taught in English. The program intently and substantially crosses the boundaries of natural and social sciences, within the context of natural scientific innovation. The learning process is structured and driven by developing scientific research projects in the specific empirical setting. This takes place within the following subject areas:

1) Energy and Sustainability, with an emphasis on renewable and clean energy development;

2) Life and Health, emphasizing drug development, molecular diagnostics and innovative medical instrumentation.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Science, Business & Innovation Year 1

Courses

Course Name	Period	Credits	Code
Science Project	P4+5	18EC	XM_0180

Programme Sections

- SBI Compulsory Courses Year 1

SBI Compulsory Courses Year 1

Courses

Course Name	Period	Credits	Code
BioSolar Cells	P1	6EC	X_428531
Business, Innovation and Value Creation in the Life Science Industry	P1	6EC	X_432723
Digital Innovation: New Ways of Organizing and Working	P2	6EC	X_400653
Management of Sustainable Innovation	P2	6EC	E_DBI_MANSI
Current Sustainable Energy Technologies	P3	6EC	X_422582
Science Project	P4+5	18EC	XM_0180
Organizing in a Digital Society	P5	6EC	E_BA_ODS
Project Sustainable Ecosystems	P6	6EC	XM_0145

Science, Business & Innovation Year 2

Programme Sections

- Science, Business and Innovation specialisation Energy and Sustainability
- Science, Business and Innovation specialisation Life and Health

Science, Business and Innovation specialisation Energy and Sustainability

Programme Sections

- Science, Business and Innovation constrained choice
- Science, Business and Innovation constrained choice - Energy and Sustainability
- Science, Business and Innovation electives
- Science, Business and Innovation compulsory courses

Science, Business and Innovation constrained choice

Description

One specialisation-related compulsory integration course (6 EC). Students who are following the Energy & Sustainability specialisation can follow the course: Current Sustainable Energy Technologies (X_422582). Students who are following the Life & Health specialisation can follow the course Business, Innovation and Value Creation in the Life Science Industry (X_432723).

Student can choose one or the other but have to complete one of both.

Courses

Course Name	Period	Credits	Code
Business, Innovation and Value Creation in the Life Science Industry	P1	6EC	X_432723
Current Sustainable Energy Technologies	P3	6EC	X_422582

Science, Business and Innovation constrained choice - Energy and Sustainability

Description

Students register for the following two courses:

- Biosolar Cells (6 EC)
- Project Sustainable Ecosystems (6 EC)

If Green and Circular Chemistry (XMU_0040) is obtained in academic year 2024-2025 or earlier, choose one of the two courses above.

Students who start the Master programme in 2025-2026 choose the following 2 courses: Biosolar Cells (6 EC) and Project Sustainable Ecosystems (6 EC).

Courses

Course Name	Period	Credits	Code
BioSolar Cells	P1	6EC	X_428531
Project Sustainable Ecosystems	P6	6EC	XM_0145

Science, Business and Innovation electives

Description

Students follow 18 EC of electives. Students can follow courses from other master programs at the VU, or from other universities in the Netherlands or abroad. Selected courses must first be approved by the Examination Board.

Courses

Course Name	Period	Credits	Code
Management and Practice of Consulting	P1	6EC	E_BA_MPC
Management of Digital Innovation	P1	6EC	E_DBI_MDI
Strategy Evaluation and Advice	P1	6EC	E_BA_SEA
The Entrepreneurial Venture	P1	6EC	E_ENT_TEV
Environmental Policy	P2	6EC	AM_468021
New Business Models and Consulting	P2	6EC	E_BA_NBMC
System innovations and sustainable transitions	P2	6EC	XM_0144
International Intellectual Property Law	P4	6EC	R_Int.prop.l
Science and Society in Historical Perspective	P4+5	6EC	X_400424
Organizing in a Digital Society	P5	6EC	E_BA_ODS

Science, Business and Innovation compulsory courses

Description

In the first year, period 1 all students must have followed the 6 EC compulsory course SBI Research Methodology. All students must follow 18 EC of compulsory courses in business and innovation. All students finish the programme with a 36 EC SBI project and Master Thesis.

Courses

Course Name	Period	Credits	Code
SBI Project & Master Thesis	Ac. Year (September)	36EC	X_432735
Digital Innovation: New Ways of Organizing and Working	P2	6EC	X_400653
Management of Sustainable Innovation	P2	6EC	E_DBI_MANSI

Science, Business and Innovation specialisation Life and Health

Programme Sections

- Science, Business and Innovation constrained choice
- Science, Business and Innovation electives
- Science, Business and Innovation constrained choice - Life and Health
- Science, Business and Innovation compulsory courses

Science, Business and Innovation constrained choice

Description

One specialisation-related compulsory integration course (6 EC). Students who are following the Energy & Sustainability specialisation can follow the course: Current Sustainable Energy Technologies (X_422582). Students who are following the Life & Health specialisation can follow the course Business, Innovation and Value Creation in the Life Science Industry (X_432723).

Student can choose one or the other but have to complete one of both.

Courses

Course Name	Period	Credits	Code
Business, Innovation and Value Creation in the Life Science Industry	P1	6EC	X_432723
Current Sustainable Energy Technologies	P3	6EC	X_422582

Science, Business and Innovation electives

Description

Students follow 18 EC of electives. Students can follow courses from other master programs at the VU, or from other universities in the Netherlands or abroad. Selected courses must first be approved by the Examination Board.

Courses

Course Name	Period	Credits	Code
Management and Practice of Consulting	P1	6EC	E_BA_MPC
Management of Digital Innovation	P1	6EC	E_DBI_MDI
Strategy Evaluation and Advice	P1	6EC	E_BA_SEA
The Entrepreneurial Venture	P1	6EC	E_ENT_TEV
Environmental Policy	P2	6EC	AM_468021
New Business Models and Consulting	P2	6EC	E_BA_NBMC
System innovations and sustainable transitions	P2	6EC	XM_0144
International Intellectual Property Law	P4	6EC	R_Int.prop.l
Science and Society in Historical Perspective	P4+5	6EC	X_400424
Organizing in a Digital Society	P5	6EC	E_BA_ODS

Science, Business and Innovation constrained choice - Life and Health

Description

Students choose 2 courses from the following list:

- Caput Protein Structure Function and Valorization (6 EC)
- From Physics to Physiology (6 EC)
- Principles of Drug Targets (6 EC)
- Principles of Drug Discovery (6 EC)

If Green and Circular Chemistry (XMU_0040) is obtained in academic year 2024-2025 or earlier, choose one of the courses above. Students who start the Master programme in 2025-2026 choose 2 courses from the following list: Caput Protein Structure Function and Valorization (6 EC), From Physics to Physiology (6 EC), Principles of Drug Targets (6 EC) or Principles of Drug Discovery (6 EC).

Courses

Course Name	Period	Credits	Code
From Physics to Physiology	P1	6EC	X_430112

Course Name	Period	Credits	Code
Principles of Drug Targets	P1	6EC	XM_0105
Principles of Drug Discovery	P2	6EC	XM_0106
Caput Protein Structure, Function and Valorization	P4+5+6	6EC	XM_0133

Science, Business and Innovation compulsory courses

Description

In the first year, period 1 all students must have followed the 6 EC compulsory course SBI Research Methodology. All students must follow 18 EC of compulsory courses in business and innovation. All students finish the programme with a 36 EC SBI project and Master Thesis.

Courses

Course Name	Period	Credits	Code
SBI Project & Master Thesis	Ac. Year (September)	36EC	X_432735
Digital Innovation: New Ways of Organizing and Working	P2	6EC	X_400653
Management of Sustainable Innovation	P2	6EC	E_DBI_MANSI

Courses

BioSolar Cells

Course Code	X_428531
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.N. Frese
Examiner	dr. R.N. Frese
Teaching Staff	dr. R.N. Frese
Teaching method(s)	Lecture

Course Objective

The course provides a broad overview of BioSolar Cells scientific research. It serves to acquire detailed knowledge of parts that comprise biological, chemical and physical materials approaches to utilize sunlight for fuel and or capture carbon dioxide based on biological or artificial photosynthesis.

Focus is on technologies that are being researched right now and are not yet implemented at full scale. Part of the course is dedicated to technology transition theory, focusing on the hurdles and bottlenecks for lab-based researched to be implemented. Important skills that are developed are the assessment of technologies that are not yet implemented at scale for current decision makers in renewable energy.

In the course, students will be reading scientific papers from the exact sciences and other academics, information leaflets for different target audiences on renewable energy efforts. Subsequently, students will be following lectures on focused topics and learn to distill relevant information from them. In work, students aim to present an overview of specific technology using scientific and other references with an assessment of the relevance of references used.

Combined this course will allow students to investigate a topic of their own interest and to acquire the necessary skills of assessing and presenting scientific information. The course provides the basic skills for a successful master study program.

The course consists of

- a group assignment on a relevant issue related to Biosolar Cells, presentation
- a literature study on one of the themes from BioSolar Cells, using a template geared to fast and critical

information transfer
- a set of questions and annotations in a diverse set of reading materials

Course Content

The course consists of three parts, and starts with interactive lectures by experts on each of the three themes from BioSolar Cells. Students will read one or two papers before each lecture and formulate questions that have been researched by themselves first. In the second stage of this course, the students first work out a group assignment (groups of 3-4 students) leading to a team presentation on a specific topic, and then conduct an individual assessment on a BioSolar-Cells related topic of their own choice and present their work in the form of a scientific review paper where the specific topic is compared with other solutions, the paper concludes with a brief description of a research proposal.

Additional Information Teaching Methods

- Lecture
- Self-study

Method of Assessment

Grade is a weighted sum of the presentation and the report.

For the quality of the questions submitted prior to the lectures the student will receive the judgement “insufficient”, “sufficient” or “excellent” by the lecturer and a +/- bonus will be added/subtracted from the grade depending on the final sum.

Also for the quality of the referee reports the student will receive the judgement “insufficient”, “sufficient” or “excellent” and +/- bonus point. Maximum grade is always 10.

A retake is only allowed once and only when all materials have been submitted with requested content and within the prescribed deadline.

Entry Requirements

Physics or chemistry at bachelor level or a mixture of these at bachelor level supplemented with other topics. Highschool biology is recommended. For those without prior knowledge on photosynthesis an introductory lecture and reading material will be provided.

Literature

During the course a diverse set of scientific papers, leaflets, academic papers, brochures, and other materials is provided for reading and annotating (compulsory)

Additional Information Target Audience

master students physics, chemistry, SBI, lifesciences

Additional Information

This course can be challenging for those that do not read much.

Explanation Canvas

Information, materials, rubrics and templates are on canvas, provided on a weekly basis. Students are expected to read all materials without specific instructions from a teacher.

Recommended background knowledge

for those without prior knowledge on photosynthesis an introductory lecture will be given

Business, Innovation and Value Creation in the Life Science Industry

Course Code	X_432723
Credits	6

Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Hasanefendic
Examiner	dr. S. Hasanefendic
Teaching Staff	dr. S. Hasanefendic
Teaching method(s)	Lecture

Course Objective

Business, Innovation and Value Creation aims to meet two distinct teaching and learning goals:

a. To obtain in-depth and comprehensive insight into current science and tech innovation and entrepreneurship trends,and processes through literature and case analysis.

b. To critically utilize and apply insights and experiences gained under a. in a personal entrepreneurship and tech/science business plan (final project) in which a student selects a product or service and develops a conceptually sound, theoretically underpinned and financially viable business model for this product or service and a business strategy according to the elements learned in class.

Essential parts of this process include: building tech or science strategy, business modeling, transactional modelling, building a value proposition, assessing investment requirements, leveraging IP, marketing and commercialization planning.

Course Content

After completing this course, you will have the following skills and tools:

- understand the basic concepts and theories underlying tech development business model, strategy development, commercialization processes and sustainable innovation (sciences or technology).
- be able to explain or reflect on mechanisms of innovation in science and technology through the theoretical lens of theories on business model, strategic management and demonstration projects seminal model.
- be able to explain the functions and formulate value creation and value capture of science and tech innovation.
- be able to compare, contrast and analyze feasibility for science and tech innovation across different countries, and in different organizations for different products and processes using seminal demonstrations project model.
- be able to recognise potential of AI for promoting sustainable innovation in science and tech innovation.
- possess a know-how to commercialize new science and tech discoveries into products and services to improve wellbeing for humans.

Additional Information Teaching Methods

A mix of: a) weekly VU campus lectures and group work within a company b) Harvard Business Case (HBS) practice lectures c) Individual coaching on the business planning exercise. Outputs include a business plan and oral presentations.

Method of Assessment

In order to receive 6 credits for this course, each student must meet the following criteria:

Be present during lectures, group work in the company and all HBS case lectures. Absence may result in a lower final grade.

Business plan: 60%
Presentation and slidedeck: 40%

You have to score at least 5.5 in all segments of evaluation-homework assignments and final project to pass this course.

Entry Requirements

Completed Bachelor SBI or comparible

Literature

Main books and worksheets (will provide relevant chapters):

- Aulet, B. (2013). *Disciplined entrepreneurship: 24 steps to a successful startup*. John Wiley & Sons.

For the upscaling tech model:

- Hasanefendic, S., Hoogstraaten, M., Bloemendal, M., Boon, W., Brezet, H., Chappin, M. M., ... & Bossink, B. (2025). Sustainable energy experiments and demonstrations: Reviewing research, market and societal trends. Energy Research & Social Science, 122, 104018.
- Bossink, B., Blankesteijn, M. L., & Hasanefendic, S. (2023). Upscaling sustainable energy technology: From demonstration to transformation. Energy Research & Social Science, 103, 103208.
- Bossink, B. (2020). Learning strategies in sustainable energy demonstration projects: What organizations learn from sustainable energy demonstrations. *Renewable and Sustainable Energy Reviews*, *131*, 110025.

Selected articles:

- Bennani, Y. L. (2011). Drug discovery in the next decade: innovation needed ASAP. Drug Discovery Today, 16(17-18), 779-792.
- Bunin, B. A., & Ekins, S. (2011). Alternative business models for drug discovery.
- Chesbrough, Henry W. "The era of open innovation." MIT Sloan Management Review, vol. 44, no. 3, 2003, p. 35+.
- Von Hippel, E. (2009). Democratizing innovation: the evolving phenomenon of user innovation. International Journal of Innovation Science.

Additional Information Target Audience

Ma SBI, Ma MPA, Ma Oncology

Additional Information

Please note that the course is highly intensive, group work is expected and immersion into a company.

Explanation Canvas

We will use Canvas

Caput Protein Structure, Function and Valorization

Course Code	XM_0133
Credits	6
Period	P4+5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. Drienovska
Examiner	dr. I. Drienovska
Teaching Staff	
Teaching method(s)	

Course Objective

For the student to develop insight into the relation between protein structure and (mal-) function.The student learns to screen and evaluate scientific literature and present a structured review of recent advances in a relevant field/topic, using the correct terminology.

Course Content

Students will perform a literature review to gain insight into the structure - function aspects of a given protein.

This insight can be applied in various areas of fundamental scientific research on protein structures such as (but not limited to): 1. protein (dis)function in disease and corresponding drug development or 2. the industrial application of enzymes.

Students investigate the valorization potential by investigating literature on drug development or (potential) industrial applications.

Additional Information Teaching Methods

Introduction to the topic will be in the form of (recorded) lectures by various lecturers. The student will select a topic and provide a brief plan of action to the relevant lecturer. Alternatively, the student receives several original publications on a topic in protein structure/disease (see above) from the lecturer. The student will study these papers and collect more information (data-base search etc.) about research in the field and then provide a plan of action for the remainder of the review. The lecturer will provide feedback on a plan of action and on a preliminary draft of the review. Finally, the student will write a report in a review-style paper discussing at least 50 references relevant to the topic.

Method of Assessment

The student write a literature review and are assessed based on the process of preparing and writing it in 4 assignments:

- Assignment 1: “Justification of topic choice”: is weighted 20% of the final grade,
- Assignment 2: “Rough outline & literature search”: is weighted 25% of the final grade,
- Assignment 3: “Detailed outline and fine structure of subchapters”: is weighted 25% of the final grade,
- Assignment 4: “Final Review” 30%: is weighted of the final grade

All assignments are graded using Rubics.

For a successful participation in this caput each assignment needs to be passed with ≥5.5. The final grade is rounded up or down to 0.5 grading points. A 5.5 is rounded up to a 6.0.

There will be one re-sit option for a failed assignment throughout the whole process.

Entry Requirements

This caput builds up on basic knowledge of proteins and their cellular functions. Successful completion of a SBI BSc and Chemical Biology, Principles of Pharmaceutical Sciences or a comparable Masters-level course on proteins is expected. Alternatively, a successful completion of a FAR BSc degree is expected.

Literature

Literature depends on the topic chosen by the student. Literature search in self-study.

Additional Information Target Audience

Masters students Science, Business and Innovation (track Life and Health)

Additional Information

This caput can be completed in the period from April to July.

This course is allowing the use of generative AI based language models (chatGPT) and teaches an effective and ethical usage of these kind of tools.

Current Sustainable Energy Technologies

Course Code	X_422582
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.N. Frese
Examiner	dr. R.N. Frese
Teaching Staff	dr. R.N. Frese, dr. A. Baldi
Teaching method(s)	Seminar, Lecture

Course Objective

This course is about asking the right questions and obtaining reliable answers on everything related to energy usage and existing, large scale, renewable energy resources. You will read the book Renewable energy without the

hot air, which provides the starting point of the energy transition we are currently bringing about. After reading and commenting on the text, you can explain the basic features of these technologies and make quantitative predictions for implementing each of these technologies. During the course, you will investigate topics of your own choice, focus on their current status, and seek ways to remove hurdles and bottlenecks. After the course, you have an overview on how to investigate the technical, economic and societal feasibilities of all possible forms of sustainable energy, including relevance and positive and negative effects.

Course Content

The course deals with all parts, Part I, II and III, of the book of MacKay, Renewable energy without the hot air, which is free to download at the book website. During the course you read the book of MacKay and annotate and discuss parts with fellow students.

Based on this and your own interests, you focus on the current state of specific topics at general media websites and other sources, from which you formulate several questions. You research topics of interest first, and focus on specific aspects that cannot be found online and are of interest to policymakers or others involved in renewable energy. From this brief investigation, you formulate focused questions with impact and provide a hypothetical answer with only rough calculations. The hypothetical answers should contain well-argued estimates. All questions must be relevant and realistic and will as such be evaluated.

During several workgroups you will learn as a team the methodology and style of reporting, and receive feedback. Subsequently. teams present their findings and you continue to work individually on your own questions including some from a fellow student. Towards the end you will review the written work of another student, and after the feedback on your work is received, you take the comments into account and prepare the final version of your report.

Attendance to all lectures and workgroups is compulsory. All work must be submitted completely and before the deadlines in order to pass this course or to be eligible for a retake. A minimum score applies for parts of the work in order to pass the course. No parts of the course are transferrable to another year.

Additional Information Teaching Methods

This course is both team and self study based, with feedback during workgroups from students and teachers. You are expected to research topics via sources such as general news and media sites, dedicated websites for specific technologies and expert reports on implementation of technology at a specific place. You report multiple times on your research and can discuss impact of your research within workgroups and individually with teachers. An active study approach of an individual student towards asking questions and feedback is essential.

Method of Assessment

The grade is a weighted average of the following items:

1. Formulation of topical researched questions and hypothesis
2. Answers to questions using the correct procedure
3. reading the book and annotating and discussing topics
4. reviewing work of fellow students

A retake is only allowed once, and only for students that have:

1. participated in all workgroup sessions
2. read the entire book with the minimal requested annotations
3. submitted all requested questions and all answers within the deadline with relevant content
4. submitted a review containing relevant content

Note: no parts of this course is transferable to another year

Entry Requirements

This course is for energy related master programs only. Prior knowledge that is expected is general knowledge of energy and power generation, renewable energy and related issues globally. This course necessitate skills like finding reliable information on the internet, critically assessing claims related to energy and output on websites or other sources. Also students should be able to assess reliable news sites for topical aspects, quick assessing data for relevance and quick reading of reports by municipalities or organizations.

The course is open to all master students within a Chemistry, Physics or Science Business and Innovation program . As well as all students with a bachelor degree in any of these three programs. All other students must have successfully followed courses in mechanics and electricity and magnetism at university level in possession of a physics based working knowledge of energy and energy related topics.

Literature

Literature: David J.C. Mackay, 'Sustainable energy - without the hot air', 2008; available free online at ttp://www.withouthotair.com
Prior the course you can browse the book but do not read it as this is being monitored during the course and part of the grade.

Additional literature that you can read: general media websites on renewable energy, news sites, sustainability and repositories of energy data

Additional Information Target Audience

This is a full time course. Following other courses, including internship in the same period is not recommended. Note: work or parts of work cannot be transferred to another year.

Additional Information

This course is individual research driven and contains reading of the book, and reading of topics of interest on the internet or other sources. Subsequent self-formulating research questions, hypothesis and answers. Each week there are two days of three hour classes.

Explanation Canvas

On canvas most relevant information can be found in the form of examples, presentations and other docs. Students are expected to read carefully all these docs without specific instruction by a teacher. Also teachers will gradually update the canvas site, communicate to all students, specific groups or individual students. All participants are expected to actively check for messages during the course on a daily basis. Also, students can actively engage with teachers via Canvas, including continuous feedback during the course

Recommended background knowledge

This course is challenging for those that do not read much, especially no news sites, and who do not have a general working knowledge on issues related to renewable energy and sustainability in our present time. Also an active attitude towards requesting feedback in workgroups and general classes for discussion is very helpful.

Digital Innovation: New Ways of Organizing and Working

Course Code	X_400653
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.W. Hafermalz
Examiner	dr. E.W. Hafermalz
Teaching Staff	K.C. Cepa, dr. E.W. Hafermalz
Teaching method(s)	Seminar, Lecture

Course Objective

The aim of this course is to provide students with a thorough academic understanding of various contemporary theoretical perspectives on technological innovation in organisations, in particular related to digital technologies.

After successful completion, students will have:

- Acquired a basic understanding of issues around organisational theories on digital technology and innovation;
- Developed an understanding of how the nature of digital technologies affect innovation processes and collaboration;
- Developed reflective and critical skills in understanding the role of (digital) technology in organisations;
- Developed an understanding of the patterns and dynamics of innovation, as well as its protection, and connected ecosystems of actors that develop them.
- Developed an understanding of the importance of a socio-technical perspective in analysing the use of technology.

Course Content

In this course, we introduce students to current theories and perspectives on digitalization and their consequences for various aspects of organizing and strategizing. The lectures are structured in such a way that we first focus on understanding the fundamental theories of digital innovation. We then discuss implications of digitalization for processes and practices of organizing, including, for example, new forms of organizing or the implications of digitalization and e.g. GenAI for knowledge work. In the last weeks of the course we introduce you to a more critical perspective on digital innovation, where we also reflect on unintended consequences of digitalization for work and society. You will learn to approach questions and phenomena of digital innovation equipped with theory and practical experience. By understanding how digital innovation triggers new ways of working and organizing, you become academically prepared to understand and support the design, introduction and use of digital innovation in practice.

The course will start with the basics of 1) what is digital technology and how it affects strategizing and organizational processes 2) what digitalization triggers in terms of business model innovation and 3) what new ways of working and organizational forms are emerging as a result of digitalization. Subsequently, we move to a level of work and discuss the impact of digitalization on work practices, including effects on organizational boundaries (e.g. open innovation), knowledge work (e.g. artificial intelligence supporting work processes) and ethics (e.g. people analytics and workplace surveillance).

Additional Information Teaching Methods

The course consists of seven lectures, group assignments, and a written exam. You are expected to come to each lecture having read the required literature. Required literature includes academic articles and case materials. The purpose of the assignments is to practice using theory and frameworks that we introduce in the lectures, on cases. The assignments usually introduce a particular organisational problem and require students to use the week's readings to address it. Every week also has a seminar on the topic that is covered in the lecture. During the seminars we discuss cases, apply theoretical concepts, and/or work on the assignments. Seminars feature a diversity of interactive formats, ranging from discussions, presentations, and assignment work.

Method of Assessment

Individual exam: 55%
The aim of the written exam is to allow students to enhance their understanding of the key concepts and theories of the course and further develop their skills and knowledge. The exam will be based on the content of lectures and all assigned literature.

The exam will consist of only open questions, which will either offer a mini-case to be interpreted with the help of theoretical material of the course, or will ask students to provide their own examples to illustrate a concept or framework introduced in the literature.

Group assignments: 45%
There will be group assignments throughout the course. Assignments are aimed to practice applying theoretical material that is introduced in the lectures to cases. To successfully perform the assignments, it is therefore essential to participate actively in the interactive lectures and carefully and engage with the assigned literature on a timely basis. Grading rubrics for the assignments will be provided on canvas.

In-class mini-essays (individual)
We will have 3 in-class essay questions, spread throughout the weeks, as a way to mutually monitor understanding and encourage course participation. Format-wise, students should expect to be asked to write a short reflection piece (around 300 words) in response to a question that links directly to the readings and examples discussed in the lectures and tutorials. For example, you may be asked to explain in your own words what “generativity of digital innovation” refers to and provide a real-life example of it. The mini-essays are assessed on a “pass” or “fail” basis. Essays mostly serve a function of exam preparation, since the format is identical to the exam. In addition, they help students to self-check their ongoing understanding of the literature and spot problems early on. You need to pass 2 out of 3 mini-essays.

In order to pass the course, you need:

- A minimum of 5,5 as a total course grade.
- The grade for each assignment and the exam must be at least (not lower than) 5,0
- Pass at least 2 out of 3 mini-essays

Environmental Policy

Course Code	AM_468021
Credits	6
Period	P2

Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Blondeel
Examiner	dr. M.C. Blondeel
Teaching Staff	dr. O.E. Widerberg, C.E. Rappel
Teaching method(s)	Seminar, Lecture

Course Objective

The Environmental Policy course introduces students to the actors, institutions and instruments relevant for designing and implementing environmental policies at the local, national and supranational levels. It also introduces various evaluation techniques for analyzing the environmental policy in an increasingly complex world.

After this course, students should be able to:

1. Describe the public and private actors and institutions involved in environmental governance at local, national and supranational levels.
2. Evaluate the effectiveness and legitimacy of environmental governance at local, national and supranational levels.
3. Explain, analyze and propose solutions regarding the complexity of decision-making structures, procedures and negotiations of environmental issues.
4. Communicate the acquired knowledge from the course on the basis of a video assignment and essays.

Course Content

This course introduces environmental policy from a political science, public policy, and international relations perspective. Environmental policy is understood in terms of governance, going beyond rules and regulations by national governments. Students will learn the basic theories and concepts in environmental policy at the local, national and international level. It also discusses interaction between the various levels of governance. Besides theory and concepts, the course provides students with practical analytical tools for analyzing the effectiveness and legitimacy of environmental policies that can be applied in as well as outside academia.

The course is structured around three modes of governance: Hierarchy (command-and-control instruments, primarily carried out by public or governmental actors); Markets (market-based governance, primarily market-based instruments); and Networks (the functioning and effects of partnerships, labeling initiatives, and actor networks (with a focus on non-state actors)).

The final week is dedicated to an emerging topic in environmental policy that changes every year, for instance, environmental activism, animal rights, transformations, AI, or degrowth.

Additional Information Teaching Methods

The course is worth 6 ECTS points which corresponds to 150 hours of work per student. The course is based on a series of lectures, exercises and debates.

Method of Assessment

1. Group video assignment (40%)
2. Individual papers (60 %)

Literature

Please see syllabus on Canvas.

Additional Information Target Audience

Master students

From Physics to Physiology

Course Code	X_430112
Credits	6
Period	P1

Course Level	400

Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I.H.M. van Stokkum
Examiner	dr. I.H.M. van Stokkum
Teaching Staff	dr. I.H.M. van Stokkum
Teaching method(s)	Seminar, Study Group, Practical, Written partial exam, Lecture

Course Objective

- To understand fundamental (bio)physical processes and their relevance for physiology.
- To gain knowledge of the application of physical methods in modelling physiological processes
- To gain knowledge of the most important theoretical and practical concepts in modelling and simulation of biomedical processes at different scales.
- To gain experience with and to apply MatLab and Mathematica to acquire, analyse and evaluate biomedical signals and to model and simulate biomedical processes.

Course Content

The course starts with a general overview of the various types of models used to describe biomedical processes by parametric and non-parametric models using linear and non-linear (differential) equations. Some of the fundamental biophysical principles that have broad relevance in biomedical research will be presented. Practical problems discussed during the lectures will concentrate on modelling the cardiovascular system using mechanical and electrical analogues.

The lectures during the first four weeks will be combined and followed by working groups in which, through relatively simple exercises, experience will be gained of MatLab and Mathematica, two frequently used 4th generation computer languages.
Finally, students will be offered a choice of 1 out of several modelling problems to be solved in small groups, guided by a supervisor. At the end of the course each group will present and discuss their work with all participants and supervisors of the course.

Additional Information Teaching Methods

Lectures, working groups, assignments.

Method of Assessment

Homework assignments (Pass/Fail), report and presentation on modelling problem (40%) and written exam (60%).

Literature

Syllabus.

Additional Information Target Audience

mBMTP, mPhys, mSBI, mComputationalScience

International Intellectual Property Law

Course Code	R_Int.prop.l
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Law
Course Coordinator	prof. dr. S.J. van Gompel
Examiner	prof. dr. S.J. van Gompel
Teaching Staff	dr. mr. D.J. Gervais LLM, mr. M.G. Hoenink, prof. dr. S.J. van Gompel
Teaching method(s)	Seminar, Lecture

Course Objective

The course provides an overview of the international legal framework for the protection of intellectual property (IP). Participants will acquire the following knowledge and skills:

- a general understanding of the role of IP protection in international trade, underlying policy considerations and the current debate on imbalances in the international protection system;
- the ability to identify and solve problems arising from the international nature of intellectual property, and the ability to assess the current IP system critically in the light of social, cultural and economic needs in developing and developed countries;
- insight into the relationship between the main areas of international IP law and general questions of business law, in particular the dynamics of international trade negotiations and their impact on IP policy making;
- knowledge about the actors of the international intellectual property community and how they interact with each other, while acknowledging legal and cultural differences. In particular, participants will understand the role of governments and their interaction with private companies to achieve innovation and development goals;
- a thorough understanding of the legal positions of various parties in the intellectual property arena and how the conduct of these parties influences the further development of international intellectual property law;
- analytical skills to apply acquired knowledge of IP rules to concrete problems, such as access to medecine and access to knowledge;
- the ability to analyse and assess IP literature, case law and legal and policy documents and critically reflect upon them;
- the ability to formulate an independent and well-substantiated opinion on complex IP issues and take a substantiated position within the existing debate on that issue.

Eindtermen: 1-10, 15.

Course Content

The course focuses on the principles and minimum standards of protection established in the Berne Convention for the Protection of Literary and Artistic Works, the Paris Convention for the Protection of Industrial Property and, in particular, the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). In line with the Law in Action approach, individual issues of intellectual property protection will be placed in the context of broader societal problems and concerns, such as:

- patent protection and public health, in particular the issue of access to medicine in developing countries;
- copyright protection and freedom to create, in particular the reuse of protected works in the digital environment and for creating generative AI (artificial intelligence);
- trademark protection of powerful brands and its impact on consumer choice and purchase decisions of consumers;
- current topics on the international IP agenda and underlying policy and development goals.

International intellectual property registration systems constitute an additional topic of the course. The Madrid System for the International Registration of Marks and the Patent Cooperation Treaty will be discussed.

Additional Information Teaching Methods

The course is based on weekly lectures in which the norms of international intellectual property law, underlying policies and the current debate on the further development of the law will be discussed.

In working group sessions, the students will present and discuss selected problems of current international intellectual property law. For this purpose, student presentation groups will be established at the beginning of the course. Participation in the group presentations is required and there is no possibility to do a retake of the presentation (see the information below).

Method of Assessment

Written exam (50% of final grade).

Presentation (50% of final grade): each participant of this Course will have to give an oral presentation during one of the working group sessions. For this purpose, student presentation groups (4 members) will be established at the beginning of the Course. The presentation topics and related literature will be made available on Canvas.

The mandatory group presentations for this course qualify as a practical exercise in the meaning of art. 3.5(1)(b) Onderwijs- en Examenregeling, for which no resit possibility is offered. Participation in the group presentations is therefore required, because without a mark for the group presentations, which count for 50% of the final grade, it is impossible to pass this course.

Literature

WIPO Intellectual Property Handbook - Policy, Law and Use, 2nd edition, Geneva, 2004 (reprint 2008), WIPO

Publication No. 489(E), ISBN 92-805-1291-7 available at: https://doi.org/10.34667/tind.28661

In addition to this general overview of the international framework of protection, academic articles on specific topics of international IP law will be made available to the students as reading assignments during the course. These reading materials are made available on Canvas.

Additional Information Target Audience

Apart from regular students, the course is also available for:
Students from other universities/faculties
Exchange students
Contractor (students who pay for one course)

Courses from a master at the faculty can only be taken as a secondary course if you have a diploma that gives access to the relevant master/specialization and if you are enrolled in a master.

Additional Information

CAPACITY
A maximum number of 50 students may participate in this course.

Custom Course Registration

A maximum number of 50 students may participate in this course.

Management and Practice of Consulting

Course Code	E_BA_MPC
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	dr. A. Plotnikova
Examiner	dr. A. Plotnikova
Teaching Staff	dr. A. Plotnikova, prof. dr. A.P. de Man
Teaching method(s)	Study Group, Lecture

Course Objective

By following this course, students advance their understanding of the way consultants are organised and work with their clients in order to critically reflect on the practice of consulting and make informed career choices. The course not only aims to develop strong writing skills but also to familiarize students with various communication styles which are essential for consultants to master.

Academic and Research Skills
By following this course, you will be able to:

- Use literature to explain how consultants work and how consultancy is organised as a business and as a profession;
- Use the literature to identify and critically assess different approaches to consulting work;

Knowledge of Theory + Methods and Solving Complex Problems
By following this course, you will be able to:

- Use theoretical lenses and scientific insights to better understand, frame and act on complex and unstructured questions;

Professional Social Skills
By following this course, you will be able to:

- Get across your ideas by adapting your communication style to your audience;

Self-Reflective Professional
By following this course, you will be able to:

- Take charge of your learning path and make informed decisions about your career;

Course Content

This course will introduce you to the field of consulting, with a specific focus on how consultants are organised and what they do. We will cover topics such as:

- History of consulting and current trends like digitalisation
- The organisation of consulting and other professional firms
- The consultancy process
- The roles of consultants
- Careers in consulting

In the seminars, you will experiment with argumentation and communication styles central to the practice of consulting (writing a journal article, preparing arguments, presentations for clients, etc.)

Additional Information Teaching Methods

Lectures (4h a week), seminars (2h a week, mandatory)

Method of Assessment

Individual/group assignments, exam

Literature

Schein, E.H. 2009. Helping: How to offer, give, and receive help. San Francisco: Berret Koehler Publishers. ISBN: 157675863X (or paperback version). Articles will be announced in the course manual.

Management of Digital Innovation

Course Code	E_DBI_MDI
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. dr. P.R. Tuertscher
Examiner	prof. dr. P.R. Tuertscher
Teaching Staff	prof. dr. ir. F. Deken, K.C. Cepa, prof. dr. ir. J.J. Berends, H.K. Fults, J.J.C. Christis MA, prof. dr. P.R. Tuertscher
Teaching method(s)	Study Group, Lecture

Course Objective

Academic & research skills: Be able to critically examine established and new innovation management concepts in the area of digital innovation

Knowledge: Knowledge of the technological, organizational, processual, and economic aspects of the management of digital innovation

Professional social skills: Ability to work effectively in teams and communicate on project findings

Self-awareness: Ability to reflect on the learning process itself, which enables students to take responsibility for their own learning and knowledge creation

Course Content

The digitization of products, services, and business models requires an effective approach to the management of innovation. Innovation is increasing in speed and scope and becoming more disruptive with the digitization of business. The management of digital innovation can draw upon established innovation management concepts, but some existing approaches may have limited relevance and even be counterproductive. New concepts are emerging but their reach and effectiveness are yet to be tested. Therefore this course explores innovation management for products, services, and business models that incorporate digital technologies, with simultaneous attention for technology, process, organization, and industrial dynamics.

The course will be organized around three interrelated topics: basic properties of digital innovation (e.g. layered modular architecture, digital materiality, digital platforms); industrial dynamics associated with digital innovation (industry disruption, network economics, ecosystems, and collaboration); and innovation processes for digital innovation (e.g. traditional and agile approaches, entrepreneurial approach).

The work group sessions will center around assignments that will be organized as “course research.” Students apply concepts to real-life cases and examine the validity and usefulness of existing innovation management concepts as well as new theories focused on digital innovation.

Additional Information Teaching Methods

Lectures
Workgroup sessions

Method of Assessment

Individual assignments
Group assignments
Peer evaluation
Exam

Literature

A list of academic articles will be provided via Canvas before the start of the course.

Additional Information Target Audience

This course is only open for students of the programs:
Digital Business & Innovation
Science, Business & Innovation

Recommended background knowledge

Basic knowledge of innovation management and organization studies.

Management of Sustainable Innovation

Course Code	E_DBI_MANSI
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	D. Diriker
Examiner	D. Diriker
Teaching Staff	D. Diriker
Teaching method(s)	Study Group, Lecture

Course Objective

Bridging Theory & Practice (knowledge):
Understand and critically assess theories of organizing for sustainable innovation and understand how entrepreneurs , business leaders, governments, and citizens can help to co-create a more sustainable future and how digitalization might be leveraged to support this future.

Bridging Theory and Practice (application):
Apply theoretical insights to conceive new approaches to organizing the implementation of sustainable innovations.

Professional & Social skills:
Ability to work effectively in a team. Have a good contribution, content and process wise: Pro-active teamwork, timely, friendly and constructive.

Course Content

Sustainability is now at the core of our societies and economies: governments, companies and citizens join forces

to fight global warming, plastic pollution, and other negative developments. Rapid changes due to digitalization have further unleashed new forms of organizing that enable and accelerate coordination, collaboration, and knowledge efficiencies.

Despite the promise of digitalization and sustainability imperatives for transforming business and society, a strategic focus on convergence of these imperatives as a means for organizing and accelerating the development and implementation of sustainable innovations is relatively new for both practice and scholarship.

In this course, we explore how business leaders and entrepreneurs around the world do things differently, to radically change the way they do business, what role they want to play in our societies, and leverage digital technologies to overcome barriers to sustainable development. You will apply theories on sustainable innovation, digital innovation, innovation processes to examine contemporary cases and learn how to organize for sustainable innovation by mobilizing and integrating information, knowledge and critical resources of diverse stakeholders.

Additional Information Teaching Methods

Lectures: Interactive Lectures (Online & In-person); Guest Lectures Tutorials: Interactive discussion and case work (Online & In-person)

Method of Assessment

Individual written exam.
Group assignment (exact assignment will be announced later).
(Peer) evaluation of contribution to the course and group

Entry Requirements

For the course it is required that students know:

- The concept of Triple P: Planet, people, profit.
- The Brundtland report Our common future 1987
- The 17 UN Global Sustainability goals.

Literature

Academic articles (to be announced in course manual).

All lectures will discuss 2-3 academic articles that will help students to understand sustainable innovations and the organizing processes that are necessary to create and implement these.

Book for background reading:
Senge, Peter M., Bryan Smith, Nina Kruschwitz, Joe Laur, Sara Schley, 2008, The Necessary Revolution: How Individuals and Organizations Are Working Together to Create a Sustainable World, Doubleday (or later edition)

Additional Information Target Audience

Students that want to make a change in existing companies, or co-create change in our societies. This is a course in which students are requested to participate very actively! Students that do not have sufficient time or motivation, are advised not to follow this course.

Concrete profile:

- Students that want to become business or policy consultant in sustainability.
- Students that want to work in a community or province to develop knowledge on how regions can be transformed to co-create value in business and society.
- Students that want become social entreprenuers or implement sustainable innovations in their future careers

Recommended background knowledge

Knowledge of innovation management, diffusion of innovation.
Please note: This course is not about innovation (technical) but about the management of innovation, e.g. stakeholder management, co-creation, open innovation, sustainable entrepreneurship.

New Business Models and Consulting

Course Code	E_BA_NBMC
Credits	6

Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. dr. A.P. de Man
Examiner	prof. dr. A.P. de Man
Teaching Staff	dr. L.H. Hoeksema, prof. dr. A.P. de Man
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Learning goals include:

- Structuring unstructured problems
- Understanding how tools can be useful to structure unstructured problems
- Teamwork, interview skills, learning to get access to a company
- Understand the nature of business models

Course Content

This course requires students to develop (in a group) a new business model for a real life consultant or consultancy of their choice. Based on theoretical input, students are guided through the process of business model innovation. The course focuses more on application of academic knowledge and analysis of complex problems than on theory development (even though there is quite some attention for the theory behind business models). In the course you will learn to develop a value proposition, use the business model canvas, calculate a business case and develop an implementation plan. The best models participate in a Dragon's Den and will be judged by experienced entrepreneurs.

Additional Information Teaching Methods

- Lectures (2hr per week)
- Tutorial (2hr per week; attendance obligatory)
- Work on assignments (8hr per week)
- Read literature and doing research (8hr per week)

Method of Assessment

The final grade consists of grades of various assignments, some individual and some group assignments. This includes an individual exam.

Literature

Various academic articles

Additional Information Target Audience

Students interested in Management Consulting and business model development

Additional Information

This course is challenging because it requires students to do a consulting assignment that is unstructured and that may have various right and wrong answers. Students that look for certainty, continuous guidance from the teachers and expect the teachers to tell them the right and wrong answers have mostly failed this course or found it frustrating.

Recommended background knowledge

It is highly recommended to have some knowledge about consultants and how the consultancy industry works. Without such knowledge the course will be difficult. The course should fit well for MC students, people with experience in consulting or students that followed a consulting course elswhere. Financial knowledge is helpful as students need to calculate a business case. For all students, it is recommended to at least read the next two articles before the course starts: Kipping, M. 2002. Trapped in Their Wave: The Evolution of Management Consultancies. In T. Clark & R. Fincham (Eds.), Critical Consulting: New Perspectives on the Management Advice Industry: 28-49. Oxford: Blackwell. Tavoletti, E., Kazemargi, N., Cerruti, C., Grieco, C., & Appolloni, A. 2022. Business model innovation and digital transformation in global management consulting firms. European Journal of Innovation Management, Vol. 25 No. 6, pp. 612-636. https://doi.org/10.1108/EJIM-11-2020-0443

Organizing in a Digital Society

Course Code	E_BA_ODS
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. dr. ir. J.C. van Burg
Examiner	prof. dr. ir. J.C. van Burg
Teaching Staff	prof. dr. ir. J.C. van Burg
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Advanced Academic Skills

- You will analyze digitization in organizations and society based on the core theoretical frameworks on organizational theory and change
- You will apply the latest knowledge and analytical capabilities to become a digital-savvy manager and successfully design and lead innovation processes.

Knowledge of Theory and Methods + Complex Problem Solving

- You will develop an understanding of how digital technologies transform organizations, industries, and society;
- You will recognize and understand the major digital technology-enabled business models that have emerged or changed in recent years;

Broadening your Horizon

- You will gain a complex understanding of how digital technologies shape organizational and social structures;

Reflective Professional

- You will reflect on your own use of digital technology and how it shapes your work and relationships

Course Content

This course is one of the interdisciplinary electives of the Business Administration SHRM, IM, LCM, MC, and S&O programs. It is of interest for students of the SHRM, IM, LCM, MC, and S&O programs as it offers a contemporary treatment of various applications and implications of digital technologies. The course will help students to develop a critical understanding of the opportunities and challenges that digital technologies pose for markets, organizations, and society; now and in the future.

Additional Information Teaching Methods

Plenary lectures (in person and online) and interactive workshops/tutorials.

Method of Assessment

Individual and team assessment

Entry Requirements

Recommended: Innovation Management

Literature

A combination of cases and scholarly articles. The details will be published on Canvas.

Principles of Drug Discovery

Course Code	XM_0106

Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. I.J.P. de Esch
Examiner	prof. dr. I.J.P. de Esch
Teaching Staff	prof. dr. A.D. Windhorst, prof. dr. I.J.P. de Esch, dr. M. Wijtmans
Teaching method(s)	Seminar, Lecture

Course Objective

To gain insight and hands-on experience in key concepts underlying drug discovery, the molecular modeling tools that enable drug design and the strategies to plan efficient synthesis routes for conceived ligands.

Course Content

- You will get acquainted with key aspects of small-molecule drug discovery, such as hit finding, hit exploration and structure-activity
  relationship exploration.
- You will get acquainted with computational techniques able to generate ideas for novel compounds, such as crystal structure analysis, building homology models, predicting binding poses of ligands via molecular docking, calculating binding free energy and affinity of ligands, de novo ligand generation, and pharmacophore modeling.
- You will learn the importance of synthetic feasibility as a key and integral part of the design process and of being able to define a
  synthetic pathway for the preparation of the designed compounds, because a design that cannot be synthesized is by definition a useless design. To this end, concepts covered include retrosynthesis and the incorporation of some selected moieties, such as heteroatoromatic scaffolds and known affinity-increasers.
- You will get immersed in a drug discovery case study in which you will design a potential drug and propose a feasible synthesis route for the molecule.
- You will be exposed to state-of-the-art software to guide you through your own drug discovery project, which will result in
  an article report describing the research you performed.

Additional Information Teaching Methods

Lectures, tutorials, computer practicals, self-study, open offices, assignments, peer review and work on a parallel case study.

Method of Assessment

Exam

Literature

Patrick, G., An Introduction to Medicinal Chemistry 7th ed., Oxford:
Oxford University Press. 20017, ISBN: 9780198866664

Two E-Books that are accessible through UBVU at all VU computers:

- Mason: Volume 4 of Comprehensive Medicinal Chemistry II: Computer-Assisted Drug Design (Mason (Ed.).
  http://www.sciencedirect.com/science/referenceworks/9780080450445
- Hoffmann: Elements of Synthesis Planning (Hoffmann (Ed))
  http://www.springerlink.com/content/j81646

Additional Information Target Audience

mDDS, mCh, mBMS (biological chemistry track), mSBI (life track)

The course is not accessible for people with a BSc degree in Pharmaceutical Sciences.

Recommended background knowledge

Knowledge of basic Organic Chemistry

Principles of Drug Targets

Course Code	XM_0105
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R. Leurs
Examiner	prof. dr. R. Leurs
Teaching Staff	prof. dr. R. Leurs
Teaching method(s)	Seminar, Lecture

Course Objective

To gain insight into general steps in the process of drug discovery & development, understanding of the molecular basis of target-drug interactions and the role of pharmaco- and toxicodynamics.

- You will get acquainted with the various processes essential for drug discovery & development
- You will understand the structure and function of different drug targets
- You can explain how enzymes, receptors and DNA can be targeted by drugs
- You will understand the molecular basis of target-drug interactions and how drug-target interactions can be quantified
- You will learn the importance of drug metabolism and safety assessment

Course Content

This course is designed for students with an interest in life sciences and the biotech/pharmaceutical industry but without prior education in this field. A general introduction will be given to the process of drug discovery, drug design and synthesis, drug development and drug safety assessment. Subsequently, potential drug targets, mechanisms of drug actions (including drug- receptor/enzyme interactions and dose- response relations), drug disposition (including pharmaco-/toxicokinetics) and drug toxicity will be discussed. Various modern developments and tools will be illustrated by recent applications in the field of drug discovery.

Additional Information Teaching Methods

Lectures, tutorials, self-study, assignments.

Method of Assessment

Written exam and assignment(s)

Literature

Patrick, G., An Introduction to Medicinal Chemistry 6th ed., Oxford: Oxford University Press. 20017, ISBN: 978-0-19-874969-1, mainly Part A (Drug targets) and Part B (PDPK). Next to this textbook, some primary literature will be used to illustrate certain topics.

Additional Information Target Audience

This introductory course is primarily designed for students mDDS**without** a Bsc degree in Pharmaceutical Sciences or SBI. Accessible for students from mCh, mBMS (Biological Chemistry track) or mSBI (life track) master.

The course is **not** accessible for people with a BSc degree in Pharmaceutical Sciences or SBI.

Recommended background knowledge

Knowledge of basic biochemistry and organic chemistry

Project Sustainable Ecosystems

Course Code	XM_0145
Credits	6

Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. F.M. Béen
Examiner	dr. F.M. Béen
Teaching Staff	dr. J.F. Brusselaers MA, dr. F.M. Béen, prof. dr. M.H. Lamoree, dr. S.H. Brandsma
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

- To understand the impact that anthropogenic activities, in particular, the manufacturing and emission of chemicals have (had) on human health, ecosystems, biodiversity and the environment in general through the study of a number of relevant examples;
- To understand the concept of planetary boundaries, how they can be used to operate sustainably, as well as their limitations when it comes to regulating chemicals;
- To understand how concepts like circular economy, safe and sustainable by design chemicals and materials, and recycling play a role in creating a sustainable future, as well as the challenges that these bring with them;
- Know which legislations have been put in place at EU-level and worldwide to regulate the use of chemicals and minimize their impact on human health and the environment;
- Be able to write a report evaluating the (potential) risks associated with the use of a given chemical (group) in industrial activities, its (potential) emission in the environment and sustainable solutions to mitigate these risks;
- To improve the ability to lead and work as part of a team, giving students the opportunity to work in a team and tackle interdisciplinary subjects.

Course Content

Since the mid of the 19th century, the production and use of chemicals have been increasing dramatically and nowadays play a crucial role in the functioning of our societies. In fact, the chemical industry nowadays has become one of the most important manufacturing industries from an economic perspective. From pharmaceuticals, all the way to textiles and plastics, chemicals are involved in almost everything that we do. Yet through their life cycle, from manufacturing, during usage and eventually, when they are being disposed of, chemicals can impact human health and the environment. Through voluntary or involuntary emissions in the air (e.g., atmosphere) and water (e.g., rivers), chemicals can enter the environment and influence entire ecosystems. These chemicals can eventually also affect human health as we ourselves are exposed to them through the air we breathe, the food we eat and the water we drink.

The realization that chemicals can have an impact on human and environmental well-being, has driven the adoption of a broad range of strategies to mitigate their impact, from the development of new technologies to remove these chemicals from the environment all the way to the introduction of stringent legislations to regulate their use and prevent their emissions. At the same time, human-induced climate change and overconsumption of resources have highlighted the urgent need to shift from a linear to a circular economy that focuses on sustainability and where reuse and recycling play a crucial role. However, while these are essential for the implementation of a circular society, they also bring new challenges related to the fate of chemicals of concern during recycling processes and their potential impact on humans and the environment.

The goal of this course is to provide students with a holistic understanding of issues related to chemicals and environmental pollution and how these are intertwined with circular economy and sustainability.

Additional Information Teaching Methods

This course will make use of a range of blended learning techniques. The course will be structured in three major parts: (i) lectures on a selection of relevant topics aligned with the learning goals, (ii) self-study involving various forms of (e-)learning and (iii) working groups during which students will work on a policy report and a presentation.

Goal of the working groups: Be able to produce and present a report about the sustainable use of chemicals.

By focusing on a specific group of legacy or novel industrial chemicals, students will work in groups to prepare a report that tackles issues related to the use, production and disposal of industrial chemicals. They will incorporate existing knowledge about their use, manufacturing, emissions sources and impacts on humans and the environment, as well as link these to existing regulations about sustainable chemical management principles and international legislation (see below for a more detailed outline of a suggested report). The outcomes will be summarized in a report which will be presented to the whole class.

Method of Assessment

The course assessment will involve three parts: (i) a final exam consisting of a multiple choice questionnaire; (ii) a report and (iii) a presentation based on the outcomes of the self-study and working groups.

The report and presentation given by students will be used as a formative assessment for students. The report and presentation will be evaluated by the course coordinator and the provided feedback will allow students to determine to which extent they are on track to achieve the defined learning goals. Report and presentation will be graded and will weigh in the final (summative) assessment of students together with a final assessment.

Explanation Canvas

All the information related to this course, including assignments and lectures, will be coordinated via Canvas. So please register to the course through Canvas.

SBI Project & Master Thesis

Course Code	X_432735
Credits	36
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Hasanefendic
Examiner	dr. S. Hasanefendic
Teaching Staff	J. Ge, dr. S. Hasanefendic, dr. M.L. Blankesteijn, prof. dr. ir. B.A.G. Bossink
Teaching method(s)	Lecture

Course Objective

1. Design, execute and report on research 2. Aptitude test (the thesis).
The 'aptitude test' means that the acquired knowledge, skills and attitudes in the field of Science, Business & Innovation are applied in a project. This project comprises an internship in an organization of 6 months, the formulation of an SBI-related problem, an analysis of this problem and written and oral reporting of the problem and analysis.

Course Content

When completing the Master project students learn to further deepen their knowledge and skills that they have obtained during their bachelor and master program.
At the start of the project, students develop a project plan. This plan consists of a literature study, research questions, research methods and techniques, a time schedule and research goals.
The student is only allowed to start with the project, when the plan has been approved by the supervisors from the VU University. The SBI master project (6 months, 36 EC) is a research internship.
The objectives of the SBI master project are:

1. Research, analyse and objectify a research question existing within an organization. Derive answers to the research question(s) based on results obtained during execution of the research project, discuss outcomes and derive conclusions
2. Aim to formulate recommendations to the company based on the research outcomes and conclusions
3. Gain insight into the strategy and operation of an organization
4. Practice and develop social and communication skills
5. Orientation on a future field of work.

Additional Information Teaching Methods

During the master project, three lectures and monthly consultnacy sessions will be given at the university. Each student wil have a designed first and second supervisor for feedback at regular intervals. For further information see Manual Master project SBI (Canvas). The student will spend most of his/her time on conducting the research project and writing the thesis.

Method of Assessment

Work execution: 20%
Aptitude test (the thesis): 60%
Final oral presentation: 20%

Entry Requirements

Students may participate in the SBI Project & Master Thesis (X_432735) only if they have passed the interim examination or examinations for the components mentioned hereinafter:

- SBI Research Methodology (X_432846)
- Science Project (XM_422591)

Literature

- Verschuren, P., Doorewaard, H. (most recent edition) Designing a research project. The Hague: Eleven International Publishing.
- Other literature as described in the plan of action.

Additional Information Target Audience

2 M SBI

Additional Information

Students must write a reflection report when they have (almost) completed their internship.

Science and Society in Historical Perspective

Course Code	X_400424
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G.J. Somsen
Examiner	dr. G.J. Somsen
Teaching Staff	dr. G.J. Somsen, dr. D.J. Beckers
Teaching method(s)	Seminar, Lecture

Course Objective

To understand how science and society have co-evolved and to know how to research this co-evolution.

Course Content

We are used to thinking of science as the prime shaper of our modern world, but how exactly does that work? How have the sciences come to play this role in our society and, reversely, how have politics and culture impacted the scientific enterprise? In this course we will study the changing relationship between science and society over the centuries by looking at a series of themes. We will examine how scientists have dealt with the public, we will study science’s relationship to religion, and we will ask how Western science really is. To address these questions, we will use historical literature, videos and podcasts as well as original source texts and pictures. We will also try our own hand at historical research, through a number of skills trainings and a final paper.

Additional Information Teaching Methods

Seminars (2hrs per week), skills trainings (twice), lectures (occasionally), presentations ands feedback.

Method of Assessment

Assessment is based on active participation during the seminar (10%) and a piece of historical research, reported on in a presentation (20%) and a final paper (70%).

Entry Requirements

Bachelor degree

Literature

Available via Canvas

Additional Information Target Audience

- Master students in the sciences interested in historical reflection;
- Master students in the humanities interested in understanding the role of science in society.

Additional Information

More information with the course coordinator: Geert Somsen, Athena Institute, g.j.somsen@vu.nl

Recommended background knowledge

None necessary, except what you have learned in your BA.

Science Project

Course Code	XM_0180
Credits	18
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. B.A.G. Bossink
Examiner	prof. dr. ir. B.A.G. Bossink
Teaching Staff	dr. S. Hasanefendic, prof. dr. ir. B.A.G. Bossink, dr. C. Ramanan
Teaching method(s)	Seminar, Other, Lecture

Course Objective

Semi-independently - under the supervision and guidance of several senior researchers and in consultation with a group of fellow master student-researchers - setting up, executing, and reporting (in writing and orally) of a scientific case study research project. The research project is aimed at analyzing the characteristics of the breakthrough technology and at identifying and interpreting the market and social valorization possibilities of the breakthrough technology.

Course Content

The course consists of Working Group Meetings and an Individual Research Assignment. In the meetings, which are compulsory, students are guided and supported through the complete research process. In the time between the work groups, the students work independently on their individual research project. Topics that will be addressed are: breakthrough technology, stakeholders, market-society opportunities, research questions, literature review, research structure, case study research, empirical data gathering, research plan and planning, report structure and writing, reporting and presenting skills.

Additional Information Teaching Methods

Work Group Meetings

Method of Assessment

The final grade consists of:

- Participation and attendance
- Research paper (written report, poster and poster pitch)

Strategy Evaluation and Advice

Course Code	E_BA_SEA
Credits	6
Period	P1
Course Level	400

Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	I. Chatterjee
Examiner	I. Chatterjee
Teaching Staff	I. Chatterjee, dr. M. van Grinsven
Teaching method(s)	Study Group, Lecture

Course Objective

Academic and Research Skills:
You will strengthen your critical thinking and analytical abilities by evaluating consultancy reports and making evidence-based recommendations. The course enhances your skills in researching strategic issues, structuring arguments, and communicating insights clearly and persuasively.

Knowledge of Theory + Methods and Solving Complex Problems:
You will apply strategic decision-making frameworks and communication theories to assess the strengths and limitations of existing strategies. Based on this analysis, you'll propose innovative, theory-informed solutions that respond to real-world organizational challenges.

Professional Social Skills:
The course strengthens your ability to collaborate in teams, communicate effectively, and engage with stakeholders. You develop awareness of the stakeholder context in which a consultant operates and the environmental impacts on the client.

Horizon Beyond Professional Area:
The knowledge and skills you acquire include social and team-working skills and critical thinking skills. You will also learn about the increasing use of AI in data analytics and visualization.

Self-Reflective Professional:
You learn to reflect on your work, give and receive peer feedback, and evaluate your professional growth towards becoming a consultant.

Course Content

As a practice-oriented course within the Management Consulting specialization,*Strategy Evaluation and Advice* immerses you in real-world consulting scenarios. You'll work on assignments that mirror professional practice—evaluating strategy recommendations from recent consultancy reports, engaging with stakeholders, developing analytical frameworks, and crafting persuasive reports. Consultants are frequently asked not only to advise on strategy but also to assess and validate strategies developed internally. In this course, you'll build the skills to critically evaluate strategic plans, formulate well-supported perspectives, and communicate your advice with clarity and impact.

Additional Information Teaching Methods

Lectures and tutorials

Method of Assessment

Assignments (group report), exam

Entry Requirements

none

Literature

Reading materials will be announced in the course manual.

Additional Information

None

Recommended background knowledge

None

System innovations and sustainable transitions

Course Code	XM_0144
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K.P.W. Kok
Examiner	dr. K.P.W. Kok
Teaching Staff	dr. K.P.W. Kok
Teaching method(s)	Other

Course Objective

The course aims to pursue the following learning objectives:

- to acquire knowledge on (the need for) system innovations and sustainable transitions;
- to understand and apply state-of-the-art theories and insights on understanding and governing sustainable transitions;
- to understand the roles of transdisciplinary research to study transitions;
- to assess and reflect upon wider political challenges in system innovation and sustainable transitions;
- to design and execute a collaborative research project on an empirical transition case;
- to acquire and practice skills to collect and analyze data (such as in-depth, semi-structured interviews);
- to evaluate and translate research findings into (policy) recommendations to support sustainable transitions.

Course Content

There is an urgent need to tackle a number of interlinked and persistent societal challenges, such as the climate crisis, biodiversity loss and socio-economic inequalities. Addressing such challenges requires system innovations and sustainable transitions in different societal domains (energy, health, food). Sustainability transitions are complex and large-scale processes of long-term societal change and they involve changes in all aspects of society, such as technology, policy, markets and socio-cultural values. Crucial in the acceleration of sustainability transitions is the upscaling of innovations that can help address societal challenges. However, because of their complex nature, transition processes are very difficult to ‘steer’ or ‘manage’. But how then do we make sure that we can accelerate sustainable transformation in light of urgent crises? How can we scale-up those system innovations that have broad societal support? And how can we make sure that these transitions are not only environmentally sustainable, but also lead to socially just and democratic futures across the globe?

In this course, you will learn about system innovation and sustainable transitions. You will learn about state-of-the-art insights on both understanding and steering transition processes. Based on literature and insights from ongoing transitions in e.g., food, energy, and health systems you will learn about the barriers to sustainable transformation, as well as strategies and opportunities for upscaling of innovations. Bringing transitions into practice requires transdisciplinary collaboration between actors from science and society (policy makers, innovators, NGOs). Therefore, you will learn about transdisciplinary research methods and initiatives (transition experiments, Living Labs) that try to both understand and accelerate transitions. To bring your knowledge into practice, you will design and execute a collaborative research project together with other students in an interdisciplinary group. In this project you will analyze an empirical case of a transition process, you will acquire different skills and tools to use in analyzing transitions.

Additional Information Teaching Methods

In order to achieve the learning objectives, several teaching methods are used including interactive seminars, project meetings and self-study. Please note that presence during the seminars and project meetings is mandatory.

Method of Assessment

The course grade is based on an oral exam (mid-term, individual grade) and the project (end of the course, report and presentation, group grade). All of the following need to be concluded with the grade of 5.5 or higher to pass the course.

- Individual oral exam (40%)
- Team project report (40%)
- Team project presentation (20%)

Literature

Articles and book chapters are made available via Canvas.

Additional Information Target Audience

This course is open to Master students with an interest in system innovations and sustainability transitions.

It is an elective course for students following the Master Science, Business and Innovation at the Faculty of Exact Sciences, VU Amsterdam.

Explanation Canvas

More information will be made available through Canvas.

The Entrepreneurial Venture

Course Code	E_ENT_TEV
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. dr. E. Masurel
Examiner	prof. dr. E. Masurel
Teaching Staff	dr. M. Mulhuijzen, prof. dr. E. Masurel
Teaching method(s)	Study Group, Lecture

Course Objective

The learning objectives of this course are as follows. Upon having successfully passed this course, the student:

1. Is able to make relevant contributions to the field of the entrepreneurial venture during its life cycle, using academic research skills (this learning objective is related to Exit Qualification 1 of the Master).
2. Is able to apply the acquired knowledge of the entrepreneurial venture during its life cycle for developing innovative and scalable business ideas (this learning objective is related to Exit Qualification 2 of the Master).
3. Is able to integrate knowledge on the entrepreneurial venture during its life cycle, handle complexity, and formulate judgments with incomplete information (this learning objective is related to Exit Qualification 3 of the Master).
4. Is able to clearly communicate his/her findings related to the entrepreneurial venture during its lifecycle, including the underpinning knowledge and rationale (this learning objective is related to Exit Qualification 4 of the Master).
5. Is able to balance economic, social and environmental aspects, in order to formulate judgments concerning the development of the entrepreneurial venture during its life cycle (this learning objective is related to Exit Qualification 5 of the Master).
6. Is able to study the entrepreneurial venture during its life cycle in a largely self-directed way (this learning objective is related to Exit Qualification 6 of the Master).
7. Has developed an attitude that is relevant for the development of the entrepreneurial venture during its life cycle (this learning objective is related to Exit Qualification 7 of the Master).

See also the Teaching and Examination Regulations 2025-2026 for the full description of the Exit Qualifications of the Joint VU & UvA Master Entrepreneurship.

In summary, by following this course, you will become an expert in the life cycle of the entrepreneurial venture / small firm, in terms of knowledge, skills and attitude.

Course Content

The focus of this course is on the changes of the entrepreneurial venture during its lifecycle. These changes are dealt with both in an actual way (how it is) and in a normative way (how it should be).

The ‘life cycle of the small firm’ model plays a pivotal role in this course. This is the model in which the entrepreneurial venture develops in size, through a number of different stages, when time passes. One standard is to make a distinction between the stages of conception, start-up, scale-up (or growth), maturity and decline, but other approaches are possible as well. The entrepreneurial venture is not dealt with in isolation in this course, but is

dealt with within the context of the entrepreneurial ecosystem and the business environment as well. There is also a coherence with the other course in this period (The Entrepreneurial Individual), as the entrepreneur also changes during the life cycle of the small firm. Together these courses deliver an interesting synergy, though there is clearly a difference in perspective.

The creation of entrepreneurial ventures and the further development of entrepreneurial ventures are important for the economic and societal development, especially (but not exclusively) for employment creation and for the production of (sustainable) innovations. This interest does not only apply to independent entrepreneurial ventures but it also applies to new ventures and spin-offs from large corporates, from (semi-)public institutions and from universities, and other entities.

Additional Information Teaching Methods

The teaching forms used in this course are: lectures, TEV assignment, coaching sessions (part of the TEV assignment), and weekly assignments. Within these teaching forms a great variety of approaches take place.

Method of Assessment

The assessment of the course consists of three parts: the online open book exam (60% of the final grade), the TEV assignment (40% of the final grade) and the weekly assignments (Pass/Fail). The online open book exam is an individual assignment, the TEV assignment is a group assignment, and the weekly assignments are to be made by changing duos of students.

The online open book exam tests the student’s knowledge of the literature, knowledge of the content of the lectures, ability to apply the mentioned knowledge and personal insights (mainly based on the mentioned knowledge). The TEV assignment will be explained during the second week of the course. The weekly assignments will be explained during the lectures and will be feedbacked in the class room.

Both the exam and the TEV assignment should be sufficient, i.e. completed with a grade of at least 5.5. All weekly assignments have to be completed with a Pass. All forms of examination have one re-sit.

Entry Requirements

See program entry requirements.

Literature

To be announced.